

Ryan Martin’s contribution to the Discussion of “Estimating means of bounded random variables by betting” by Ian Waudby-Smith and Aaditya Ramdas

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Congratulations to Waudby-Smith and Ramdas (WSR) for their excellent contribution to the rapidly-growing literature on anytime-valid inference. Since some of my work involves imprecise probability, Professor Ramdas asked me privately what, if any, connections there are between e-values, etc. and imprecise probability. The answer to his question might be of general interest, so I’ll share it here.

Following WSR, let $(X_t : t \geq 1)$ be a $[0, 1]$ -valued process with distribution $P \in \mathcal{P}^\mu$ having unknown mean μ . Write $X^t = (X_1, \dots, X_t)$ and x^t for a generic realization. Take $M(X^t, m)$ to be an e-value for testing $H_0^m : \mu = m$. Given x^t , define

$$\gamma_{x^t}(m) = M(x^t, m)^{-1} \wedge 1, \quad m \in [0, 1].$$

This function determines an imprecise probability for uncertainty quantification about μ , one whose upper probability is a sub-additive *possibility measure* given by

$$\bar{\Gamma}_{x^t}(A) = \sup_{m \in A} \gamma_{x^t}(m), \quad A \subseteq [0, 1].$$

$\bar{\Gamma}_{x^t}$ is a coherent upper probability, so it determines a non-empty, closed, and convex credal set $\mathcal{C}(\bar{\Gamma}_{x^t}) = \{Q : Q(\cdot) \leq \bar{\Gamma}_{x^t}(\cdot)\}$ of probability distributions dominated by $\bar{\Gamma}_{x^t}$. The credal set facilitates probabilistic uncertainty quantification about μ : a hypothesis H is discredited if $Q(H)$ is small for all $Q \in \mathcal{C}(\bar{\Gamma}_{x^t})$, i.e., if $\bar{\Gamma}_{x^t}(H)$ is small. These aren’t Bayesian-style subjective degrees of belief about μ —they’re frequentistly-justified. Indeed, WSR’s confidence sets “ C_t ” are exactly the γ -level sets: $C_t \equiv \{m : \gamma_{x^t}(m) > \alpha\}$, and there’s a uniform validity property (e.g., Cella and Martin 2023),

$$\sup_{\mu} \sup_{P \in \mathcal{P}^\mu} P \left[\bigcup_{H \subseteq [0,1]: H \ni \mu} \{X^t : \bar{\Gamma}_{X^t}(H) \leq \alpha\} \right] \leq \alpha, \quad \alpha \in [0, 1],$$

that prevents systematic discreditation of *any* true hypothesis about μ .

But there's another possibility measure that's even better than $\bar{\Gamma}_{x^t}$. Define

$$\pi_{x^t}(m) = \sup_{P \in \mathcal{P}^m} P\{M(X^t, m) \geq M(x^t, m)\}, \quad m \in [0, 1]. \quad (1)$$

If M is such that $\sup_{m \in [0, 1]} \pi_{x^t}(m) = 1$ for all x^t , then $\bar{\Pi}_{x^t}(A) = \sup_{m \in A} \pi_{x^t}(m)$ is a possibility measure. It can be shown (Martin 2022a,b) that $\bar{\Pi}_{X^t}$ enjoys the same properties as $\bar{\Gamma}_{X^t}$, but is more efficient, i.e., $\pi_{X^t}(\cdot) \leq \gamma_{X^t}(\cdot)$ and, hence, $\mathcal{C}(\bar{\Pi}_{X^t}) \subseteq \mathcal{C}(\bar{\Gamma}_{X^t})$.

For WSR's nonparametric case, computation of π_{x^t} in (1) is a challenge, so one might opt for the computationally simpler but statistically less efficient upper bound γ_{x^t} . In other cases, however, more can be done. For example, in the context of parametric models, I argued (Martin 2023) that one can efficiently balance frequentist desiderata with the likelihood principle by excluding those unrealistic and impractically extreme stopping rules, e.g., sample until the test rejects. This exclusion is achieved by suitably shrinking the domain over which an optimization like in (1) is taken.

References

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